



QUESTIONS NUMERICAL METHODS FOR WAVE PROPAGATION

DAVID FLÜCKIGER

ABSTRACT. Hier beantworten wir einige Zusammenfassende Fragen über das Fach "Numerical Methods for Wave Propagation". Diese Liste an Fragen ist keinesfalls vollständig und darf gerne weiter ergänzt werden. Die Literatur zur Vorlesung finden Sie im Literaturverzeichnis.

Question 1. Consider the one-dimensional Cauchy problem for the wave equation with constant wave speed $c > 0$. Derive d'Alembert's solution $u(t, x)$. What is the domain of dependence of a point (t_0, x_0) ?

Solution 1. Consider the Wave equation

$$(1.1) \quad \begin{cases} \partial_t^2 u(t, x) + c(x)^2 \partial_x^2 u(t, x) = 0, & x \in \mathbb{R}, t > 0 \\ u(0, \cdot) = f, & \text{on } \mathbb{R} \\ \partial_t u(0, \cdot) = g, & \text{on } \mathbb{R} \end{cases}$$

Define new variables $\xi = x + ct$, $\eta = x - ct$ and note that for $c \neq 0$ the mapping $(t, x) \mapsto (\xi, \eta)$ is bijective. Setting $v(\xi, \eta) := u(t(\xi, \eta), x(\xi, \eta))$ we obtain, using the chain rule,

$$\partial_x u(t, x) = \frac{\partial v}{\partial \xi} \frac{\partial \xi}{\partial x} + \frac{\partial v}{\partial \eta} \frac{\partial \eta}{\partial x} = v_\xi + v_\eta \quad \text{and} \quad \partial_{x,x} u(t, x) = v_{\xi,\xi} + 2v_{\xi,\eta} + v_{\eta,\eta}$$

and by a similar computation,

$$\partial_{t,t} u = c^2 v_{\xi,\xi} - 2c^2 v_{\xi,\eta} + c^2 v_{\eta,\eta}.$$

Plug this into (1.1) to deduce $v_{\xi,\eta}(\xi, \eta) = 0$ which implies after integration that

$$v(\xi, \eta) = \Phi(\xi) + \Psi(\eta) \quad \text{for some } \Phi, \Psi \in \mathcal{C}^2.$$

That is

$$u(t, x) = \Phi(x + ct) + \Psi(x - ct).$$

To find the functions Φ and Ψ we use the initial condition:

$$(1.2) \quad f(x) = \Phi(x) + \Psi(x), \quad \text{and} \quad g(x) = c(\Phi'(x) - \Psi'(x))$$

which then implies that

$$\begin{aligned} \Phi(x) &= \frac{1}{2} f(x) + \frac{1}{2c} \int_0^x g(s) ds + C_1 \\ \Psi(x) &= \frac{1}{2} f(x) - \frac{1}{2c} \int_0^x g(s) ds + C_2 \end{aligned}$$

From (1.2) we get that $C_1 + C_2 = 0$ and thus

$$u(t, x) = \frac{1}{2} [f(x + ct) + f(x - ct)] + \frac{1}{2c} \left[\int_{x-ct}^{x+ct} g(s) ds \right].$$

The domain of dependence for the wave equation is the specific region of space at the initial time ($t = 0$) that determines the state of the wave at a later space-time point (x_0, t_0) . For a 1D wave equation, it is the closed interval $[x_0 - ct_0, x_0 + ct_0]$, bounded by the characteristic lines passing through that point $\square$

Question 2. Motivate the Helmholtz equation and derive its variational and Galerkin FE formulations. Why can't we apply Lax-Milgram for existence and uniqueness? What is the Fredholm alternative? State the convergence theorem (Schatz) and explain the main steps of the proof. What is the pollution error and how can we reduce its effect?

Solution 2. Starting from the time-dependent wave equation,

$$\partial_{t,t}u - \operatorname{div}(c(x)^2 \nabla u) = f(t, x)$$

with $0 < c_{\min} \leq c(x) \leq c_{\max} < +\infty$ we now consider time time-harmonic waves,

$$u(t, x) = \operatorname{Re}(u(x)e^{i\omega t})$$

$$f(t, x) = \operatorname{Re}(f(x)e^{i\omega t})$$

governed by the single angular frequency $\omega > 0$. Then u satisfies the Helmholtz equation:

$$-\operatorname{div}(c(x)^2 \nabla u) - \omega^2 u(x) = f(x).$$

If $k(x) = \frac{\omega}{c(x)}$ is constant, then

$$(2.1) \quad \Delta u + k^2 u = \tilde{f} := -\frac{1}{c^2} f.$$

In the following, we impose homogenous Dirichlet boundary conditions. The weak formulation is, as usual, obtained by integration by parts and reads:

$$(2.2) \quad \text{Find } u \in H_0^1(\Omega) \text{ such that } a(u, v) = \ell(v) \text{ for all } v \in H_0^1(\Omega)$$

where,

$$a(u, v) = \int_{\Omega} c^2 \nabla u \cdot \nabla v - \omega^2 uv \, dx, \quad \text{and} \quad \ell(v) = \int_{\Omega} f v \, dx.$$

Note: If ω is sufficiently large, a is not coercive. This means we *cannot* apply Lax-Milgram. Actually this problem does not have a unique solution in general.

Also observe that if u_1, u_2 are solution of (2.1), then $u = u_1 - u_2$ is an eigenvector of the negative Laplacian with eigenvalue k^2 .

We need:

Theorem 2.1. Suppose that $K : H \rightarrow H$ is a compact linear operator on a real Hilbert space H . Then exactly one of the following statements is true:

- (1) For any $f \in H$ there exists a unique u such that $(\operatorname{id} - K)u = f$.
- (2) There exists $u \in H \setminus \{0\}$ such that $(\operatorname{id} - K)u = 0$.

From this we can deduce the following:

Theorem 2.2. For the weak formulation of the Helmholtz equation (2.2), precisely one of the following statements is true:

- (1) For all $f \in L^2(\Omega)$, there exists a unique solution to (2.2).
- (2) (2.2) with $f = 0$ has a non-trivial solution.

Proof. Let $f \in H := L^2(\Omega)$ and set

$$a_0(u, v) := \int_{\Omega} c^2 \nabla u \cdot \nabla v \, dx = a(u, v) - \omega^2 \langle u, v \rangle_{L^2}.$$

Since a_0 is symmetric, bilinear and coercive, by Lax-Milgramm we can define the solution operator $T : H \rightarrow H$ by

$$a_0(Tg, v) = \langle g, v \rangle_{L^2}.$$

One easily shows, using the coercivity of a_0 and $T[H] \subseteq V$, that T is bounded and linear. We now show that T is compact.

Let $\{g_n\}_{n \in \mathbb{N}}$ be a bounded sequence in H , hence $\{Tg_n\}_n$ is bounded in V . As Ω is a Lipschitz-domain, V is compactly embedded in H by Rellich, hence $\{Tg_n\}_n$ has a convergent subsequence and thus T is compact. If we set $K := \omega^2 T$ we have that K is compact and the equivalence

$$a(u, v) = \ell(v) \quad \forall v \in V \iff u - Ku = Tf$$

holds. Thus we conclude by Theorem 2.1.

Finite Elements for Helmholtz. Let $\mathcal{T}_h$ be a triangulation of Ω and define the finite elements space

$$V_h := \{v \in \mathcal{C}^0(\overline{\Omega}) \mid v|_K \in \mathcal{P}^r, K \in \mathcal{T}_h, v|_{\partial\Omega} = 0\} \subseteq H_0^1(\Omega).$$

Let $\{\varphi_i\}_{i=1}^N$ be a basis of V_h and define the matrices

$$\mathbf{A} := [a_0(\varphi_i, \varphi_j)]_{i,j=1}^N, \quad \mathbf{M} := [\langle \varphi_i, \varphi_j \rangle_{L^2}]_{i,j=1}^N, \quad \boldsymbol{\ell} := [\ell(\varphi_i)]_{i=1}^N,$$

so the discrete problems reads

$$(\mathbf{A} - \omega^2 \mathbf{M}) \mathbf{u}_h = \boldsymbol{\ell}.$$

Note that both $\mathbf{A}$ and $\mathbf{M}$ are symmetric and positive definite but $\mathbf{A} - \omega^2 \mathbf{M}$ might not be. In fact it can even be singular.

Theorem 2.3 (Schatz). *Let $\Omega \subseteq \mathbb{R}^2$ be an open convex polygon and $k > 0$ such that for every $f \in L^2(\Omega)$, (2.2) has a unique (weak) solution $u \in H_0^1(\Omega)$. In addition let $\{\mathcal{T}_h\}_h$ be a shape-regular family of triangulations of Ω . There exist constants $h_0, c_1, c_2 > 0$ such that for every $h \leq h_0$ and $f \in L^2(\Omega)$, the above Galerkin-FE formulation has a unique solution $u_h \in V_h$ and we have*

$$(2.3) \quad \|u - u_h\|_{H^1(\Omega)} \lesssim \inf_{v \in V_h} \|v - u\|_{H^1(\Omega)}$$

$$(2.4) \quad \|u - u_h\|_{L^2(\Omega)} \lesssim h \|u - u_h\|_{H^1(\Omega)}$$

Proof. By Cauchy Schwartz, a is continuous and by the usual trick one deduces Galerkin orthogonality:

$$a(u - u_h, v) = 0 \quad \text{for all } v \in V_h$$

We do not know if u_h even exists. However, since we are working in finite dimensional spaces when seeking u_h , we have

$$\text{existence and uniqueness for } f \iff \text{uniqueness for } f = 0.$$

Since for $f = 0$ we have a solution $u_h = 0$ in V_h , it will suffice to show that for h sufficiently small, this solution is unique.

Thus let us assume that u_h exists for some fixed f and h .

We now use a so-called *duality argument*: Let $w \in V$ be the solution of

$$a(w, v) = \langle u - u_h, v \rangle_{L^2} \quad \text{for all } v \in V.$$

Since Ω is convex we have $w \in H^2(\Omega)$ and

$$(2.5) \quad |w|_{H^2} \lesssim \|u - u_h\|_{L^2}.$$

By Galerkin-orthogonality, symmetry and continuity of a one finds,

$$\|u - u_h\|^2 = a(w, u - u_h) = a(w - w_h, u - u_h) \lesssim \|w - w_h\|_{H^1} \|u - u_h\|_{H^1}, \quad \text{for any } w_h \in V_h,$$

hence

$$\|u - u_h\|^2 \lesssim \|u - u_h\|_{H^1} \inf_{w_h \in V_h} \|w - w_h\|_{H^1}.$$

Using the projection error of w onto V_h we get from the last semester, using that our triangulations are shape-regular, that

$$\|u - u_h\|^2 \lesssim h \|u - u_h\|_{H^1} |w|_{H^2} \stackrel{(2.5)}{\lesssim} h \|u - u_h\|_{H^1} \|u - u_h\|_{L^2}$$

which implies (2.4).

To show (2.3) note that from the continuity of a and Galerkin orthogonality,

$$\begin{aligned} \|u - u_h\|_{H^1}^2 &= a(u - u_h, u_v) + (1 + k^2) \|u - u_h\|_{L^2}^2 \\ &\lesssim \|u - u_h\|_1 \|u - v\|_1 + \|u - u_h\|^2 \end{aligned}$$

for every $v \in V_h$, so using (2.4) one gets

$$\|u - u_h\|_1^2 \leq \hat{c} \|u - u_h\|_1 \|u - v\|_1 + \hat{c} \left(c_2 h \|u - u_h\|_1 \right)^2$$

and thus

$$\left(1 - \hat{c}(c_2 h)^2\right) \|u - u_h\|_1 \leq \hat{c} \|u - v\|_1 \quad \text{for any } v \in V_h,$$

where the constant on the LHS is bigger than 1/2 for h small enough. This implies (2.3).

If u_h exists then (2.3) and (2.4) hold. It is left to show that if h is small enough, then the discrete problem has a unique solution.

So let h be small enough and let $f = 0$. Then $u_h = 0$ satisfies the discrete problem. By (2.3), u_h is unique.

Dispersion/Pollution error see notes. □

Question 3. *Starting from a time-dependent scattering problem (including an incoming wave and an obstacle with boundary Γ) in $\mathbb{R}^d$, derive a typical time-harmonic scattering problem? What is the Sommerfeld condition? Explain in general terms how to derive a Dirichlet-to-Neumann (DtN) condition. What are the limitations of the DtN approach?*

Solution 3. Scattering problem refers to an object in a plane – or any other infinite domain Ω_∞ that has nice enough geometry – that gets hit by waves. These waves bounce off this object. The problem is, that we *cannot* simulate the whole plane, thus we solve the equation outside of the object analytically. This is possible because there we do not have any forces acting on the wave.

The idea is the following: We introduce an artificial boundary B and impose a boundary condition such that the solution stays the same in our artificial domain Ω .

Example. Consider the equation

$$\begin{cases} -u'' + k^2 u = f & \text{in } \Omega_\infty = (0, +\infty), \\ u(0) = 0, \\ \lim_{x \rightarrow +\infty} u(x) = 0, \end{cases}$$

where $f(x) = 0$ for all $x \geq 1$.¹ Let $B = \{x = L\}$ with $L > 1$. We solve the equation for $x > L$. This leads to the general solution

$$\varphi(x) = C e^{-kx} + D e^{kx}, \quad C, D \in \mathbb{R}.$$

Using the condition at $+\infty$ we deduce $D = 0$. Imposing $U_L := u(L)$ gives

$$u(x) = U_L e^{k(L-x)}, \quad \text{for } x \geq L.$$

By differentiating and evaluating at $x = L$ we get $u'(L) = -k u(L)$, and thus we get the same answer by applying the numerical method to the problem

$$\begin{cases} -u'' + k^2 u = f, & \text{in } \Omega = (0, L), \\ u(0) = 0, \\ u'(L) = -k u(L), \end{cases}$$

In general we will replace the last line by $\partial_n u(L) = \Pi u(L)$ where Π is called the *Dirichlet-to-Neumann* (DtN)-map.

Given an bounded obstacle $D \subseteq \mathbb{R}^d$ with $d \leq 3$ with $\mathcal{C}^2$ -boundary, let

$$U^{\text{tot}}(t, x) = \text{Re} \{ u^{\text{tot}}(x) e^{-i\omega t} \}$$

be a wave satisfying

$$\begin{cases} \partial_t^2 U^{\text{tot}} - c^2 \Delta U^{\text{tot}} = f & \text{in } \Omega_\infty := \mathbb{R}^d \setminus \overline{D} \\ U^{\text{tot}} = 0 & \text{on } \partial\Omega_\infty = \partial D. \end{cases}$$

We decompose U^{tot} into an incident wave U^{inc} and a scattered field U^{scat} :

$$U^{\text{tot}} = U^{\text{inc}} + U^{\text{scat}}, \quad \text{where } U^{\text{scat}}(t, x) = \text{Re} \{ u^{\text{scat}}(x) e^{-i\omega t} \}, \quad \omega > 0.$$

¹This is not the Helmholtz equation because we have $+k^2$ instead of $-k^2$.

The scattered field $u^{\text{scat}}(x)$ satisfies the Helmholtz equation in the exterior domain Ω_∞ :

$$(3.1) \quad \begin{cases} \Delta u^{\text{scat}} + k^2 u^{\text{scat}} = f, & \text{in } \Omega = (0, L), \\ u^{\text{scat}} = g, & \text{on } \partial D, \\ \lim_{r \rightarrow +\infty} r^{\frac{d-1}{2}} \left(\frac{\partial}{\partial r} u^{\text{scat}} - i k u^{\text{scat}} \right) = 0, & \text{where } r = |x|. \end{cases}$$

The last line in the previous display is the so-called *Sommerfeld's radiation condition*. We will take a closer look at this:

Example. In $d = 1$, the general solution of (3.1) is given by

$$\varphi(x) = A e^{ikx} + B e^{-ikx}, \quad A, B \in \mathbb{R}.$$

Using Sommerfeld's radiation condition gives

$$\begin{aligned} 0 &= \lim_{x \rightarrow +\infty} ((u^{\text{scat}})' - i k u^{\text{scat}}) = \lim_{x \rightarrow +\infty} \left(B(-i k e^{-ikx} - i k e^{ikx}) \right) \\ &= -2ik \lim_{x \rightarrow +\infty} B e^{ikx}, \end{aligned}$$

that is $B = 0$. The Sommerfeld selects e^{ikx} , that is, it selects only waves that travel to the right and are thus outgoing. This means there are no waves coming back from $+\infty$.

Example. In $d = 3$ we have under the assumption that u^{scat} is radially symmetric, $u^{\text{scat}} = u^{\text{scat}}(r)$,

$$0 = \Delta u^{\text{scat}} + k^2 u^{\text{scat}} = \partial_r^2 u^{\text{scat}} + \frac{2}{r} \partial_r u^{\text{scat}} + k^2 u^{\text{scat}}$$

which implies by multiplying with r^2 ,

$$0 = \frac{\partial^2}{\partial r^2} (r u^{\text{scat}}) + k^2 r u^{\text{scat}} = 0, \quad \text{that is } u^{\text{scat}} = A u_1 + B u_2, \quad \text{with } u_{1,2}(r) = \frac{1}{r} e^{\pm ikr}$$

Note that $u_{1,2}$ both go to zero as $r \rightarrow +\infty$ hence. We check that the Sommerfeld condition selects $u_1 = \frac{e^{ikr}}{r}$ and

$$U(t, r) = \text{Re} \left\{ \frac{e^{ikr}}{r} e^{-i\omega t} \right\} = \frac{\cos(kr - \omega t)}{r},$$

corresponds to purely outgoing waves. Physically this makes sense, since U^{scat} is initially zero and then generated locally near the obstacle by the incident wave. $\square$

Question 4. Derive the DtN condition (main steps only) for a 2D time-harmonic scattering problem when the artificial boundary $\mathcal{B}$ is a circle.

Solution 4. We study the problem

$$\begin{cases} \Delta u + k^2 u = 0 & \text{in } \Omega_\infty = \mathbb{R}^2 \setminus \overline{D}, \\ u = g & \text{on } \partial\Omega_\infty = \partial D, \\ \lim_{r \rightarrow +\infty} r^{1/2} \left(\frac{\partial}{\partial r} u - i k u \right) = 0, \end{cases}$$

where $D \subseteq \mathbb{R}^2$ is a bounded set with $\mathcal{C}^2$ boundary. Moreover let $\mathcal{B} := \partial B_R(0)$. In the following we will derive the DtN-map for $\mathcal{B}$.

By writing the Laplacian in Polar coordinates (r, θ) and using separation of variables and that u is smooth and 2π periodic in θ we obtain

$$u(r, \theta) = \sum_{n \geq 0} a_n \cos(n\theta) + \sum_{n \geq 1} b_n(r) \sin(n\theta),$$

where the coefficients a_n and b_n satisfy

$$\left[\partial_r^2 + \frac{1}{r} \partial_r + k^2 - \frac{n^2}{r^2} \right] c_n = 0, \quad \text{for all } r > R \text{ and } c_n \in \{a_n, b_n\}.$$

Then define $c_n = c_n(z)$ by

$$\left. \begin{matrix} a_n(r) \\ b_n(r) \end{matrix} \right\} =: c(kr) = c_n(z), \quad z = kr$$

so we obtain the Bessel equation of order n :

$$\boxed{z^2 c_n'' + z c_n' + (z^2 - n^2) c_n = 0, \quad \text{for } z > kR.}$$

These are solved by the $J_n(z)$ and $Y_n(z)$ the Bessel-function of first/second kind of order n . These satisfy

$$J_n(z) \sim \sqrt{\frac{2}{\pi z}} \cos\left(z - \frac{n\pi}{2} - \frac{\pi}{4}\right), \quad \text{and} \quad Y_n(z) \sim \sqrt{\frac{2}{\pi z}} \sin\left(z - \frac{n\pi}{2} - \frac{\pi}{4}\right), \quad \text{for } z \rightarrow +\infty$$

For the Solution we use the so-called Hankel functions of the first and second kind:

$$H_n^{(1,2)}(z) = J_n(z) \pm iY_n(z).$$

Note that the derivatives of the Hankel functions satisfy

$$\frac{d}{dz} H_n^{(1,2)}(z) = \left(\pm i - \frac{1}{2z} \right) H_n^{(1,2)}(z)$$

Then using the Sommerfeld-condition one deduces

$$\begin{aligned} \sqrt{r}(\partial_r - ik)H_n^{(1)}(kr) &\sim \sqrt{r} \left[k \left(i - \frac{1}{2kr} \right) - ik \right] H_n^{(1)}(kr) = \mathcal{O}\left(\frac{1}{r}\right), \quad r \rightarrow +\infty \\ \sqrt{r}(\partial_r - ik)H_n^{(2)}(kr) &\sim \sqrt{r} \left[k \left(-i - \frac{1}{2kr} \right) - ik \right] H_n^{(2)}(kr) \sim 2ik \sqrt{\frac{2}{\pi k}} e^{-i(kr - \frac{n\pi}{2} - \frac{\pi}{4})} \not\rightarrow 0, \\ &r \rightarrow +\infty. \end{aligned}$$

so the Sommerfeld condition selects the first Hankel-function and we get the general solution

$$u(r, \theta) = \sum_{n \geq 0} \tilde{a}_n H_n^{(1)}(kr) \cos(n\theta) + \sum_{n \geq 1} \tilde{b}_n H_n^{(1)}(kr) \sin(n\theta), \quad \tilde{a}_n, \tilde{b}_n \in \mathbb{R}.$$

Then expand $u(r, \theta)$ into its Fourier-series at $r = R$ and use the orthogonality of sine and cosine in $L^2(0, 2\pi)$ to get

$$u(r, \theta) = \frac{1}{2\pi} \sum_{n \in \mathbb{Z}} \frac{H_n^{(1)}(kr)}{H_n^{(1)}(kR)} \int_0^{2\pi} u(R, \vartheta) e^{-in(\theta - \vartheta)} d\vartheta.$$

Using trig-identities² one rewrites this as

$$u(r, \theta) = \frac{1}{\pi} \sum'_{n \geq 0} \frac{H_n^{(1)}(kr)}{H_n^{(1)}(kR)} \int_0^{2\pi} u(R, \vartheta) \cos(n(\theta - \vartheta)) d\vartheta, \quad r \geq R,$$

where we use $\sum'$ to denote a sum where the term index with 0 is multiplied with 1/2. Now take the derivative with respect to r and obtain the DtN map by setting $r = R$:

$$M[u](\theta) = \frac{k}{\pi} \sum'_{n \geq 0} \frac{(H_1^{(1)})'(kR)}{H_1^{(1)}(kR)} \int_0^{2\pi} u(R, \vartheta) \cos(n(\theta - \vartheta)) d\vartheta. \quad \square$$

Question 5. Consider the variational formulation of a 2D scattering problem with wave number $k > 0$ and a Sommerfeld-type condition

$$\frac{\partial u}{\partial n} = iku,$$

at the artificial boundary $\mathcal{B}$. Show that the solution is unique.

Solution 5. We prove the following theorem:

²Hint: use $\cos(a)\cos(b) + \sin(a)\sin(b) = \cos(a-b)$

Theorem 5.1. *The boundary value problem*

$$\begin{cases} \Delta u + k^2 u = 0 & \text{in } \Omega \\ u = g & \text{on } \Gamma \\ \frac{\partial u}{\partial n} = iku & \text{on } B \end{cases}$$

has at most one solution. Here Γ is the boundary of the obstacle D which is enclosed by Ω (but $\Omega = \Omega \setminus D$) and B is the artificial boundary around the obstacle D but is still contained in Ω .

In order to show this we need to show another theorem:

Theorem 5.2. *If $\mathcal{L} : \mathcal{C}^\infty(B) \rightarrow \mathcal{C}^\infty(B)$ is a linear (differential- or integral-) operator such that*

$$(5.1) \quad \operatorname{Im} \left\{ \int_B \bar{w} \mathcal{L}[w] \, dS \right\} < 0 \text{ (or } > 0) \text{ for all } w \in \mathcal{C}^\infty(B) \setminus \{0\}$$

then the boundary-value problem

$$\begin{cases} \Delta u + k^2 u = 0 & \text{in } \Omega \\ u = g & \text{on } \Gamma \\ \frac{\partial u}{\partial r} = \mathcal{L}[u] & \text{on } B, \end{cases}$$

has at most one classical solution.

Proof. Without loss of generality let $g = 0$. Then

$$0 = \int_{\Omega} \bar{u} (\Delta u + k^2 u) \, dx = - \int_{\Omega} |\nabla u|^2 + k^2 |u|^2 \, dx + \int_B \bar{u} \cdot \mathcal{L}[u] \, dS$$

Take the imaginary part and deduce from (5.1) that $u|_B = 0$. By writing u as Fourier series and using that the coefficients depending on r have to solve Bessels equation with zero boundary conditions we conclude that $u \equiv 0$.

Proof of Theorem 5.1. Trivially the operator $\mathcal{L}[w] = iku$ satisfies (5.1) hence the result follows. $\square$

Question 6. *Consider a 2D time-harmonic scattering problem. Derive the first-order absorbing boundary conditions $B_1 u = 0$ by Bayliss-Gunzburger-Turkel. Explain the advantages and disadvantages when compared to the DtN condition. State and prove the L^2 -error estimate between the exact solution and that using the first-order BGT-1 boundary condition at $r = R$.*

Solution 6. A typical 2D time-harmonic scattering problem is given by

$$(6.1) \quad \begin{cases} \Delta u + k^2 u = 0 & \text{for } 0 \leq r \leq R, \\ \lim_{r \rightarrow +\infty} \sqrt{r} (\partial_r u - iku) = 0 \end{cases}$$

We then know that

$$(6.2) \quad u(r, \theta) = \sum_{n \geq 0} H_n^{(1)}(kr) [a_n \cos(n\theta) + b_n \sin(n\theta)].$$

Actually we have

Lemma 6.1. *Let u satisfy (6.1). Then*

$$(6.3) \quad u(r, \theta) = H_0^{(1)}(kr) \sum_{n \geq 0} \frac{F_n(\theta)}{r^n} + H_1^{(1)}(kr) \sum_{n \geq 0} \frac{G_n(\theta)}{r^n},$$

where $F_n, G_n \in \mathcal{C}_{\text{per}}^\infty([0, 2\pi))$

Proof. A solution of (6.1) is of the form (6.2). Using the recurrence relation of Bessel functions

$$H_{n+1}^{(1)}(z) = \frac{2n}{z} H_n^{(1)}(z) - H_{n-1}^{(1)}(z), \quad n \geq 1$$

and thus using this inductively and plugging into (6.1) yields the result.

Using the asymptotic behaviour of $H_n^{(1)}$ (c.f. Question 4) we get from the above lemma

$$u(r, \theta) = e^{ikr} \sum_{n \geq 0} \frac{f_n(\theta)}{r^{n+\frac{1}{2}}}, \quad r \rightarrow +\infty.$$

Note that this is a formal power-series only, which does not need to converge at a fixed distance r . Still, by including increasingly many terms up to $n = N$, we get improved asymptotic approximations of u as $r \rightarrow +\infty$.

Idea. Find a differential operator B_m of order $m \geq 1$, such that the leading term of the asymptotic expansion of $B_m[u]$ for $r \rightarrow +\infty$ tends to 0 as quickly as possible.

Take $m = 1$ and formally differentiate the series w.r.t. r :

$$\frac{\partial}{\partial r} u(r, \theta) \sim ike^{ikr} \sum_{n \geq 0} \frac{f_n(\theta)}{r^{n+\frac{1}{2}}} - e^{ikr} \sum_{n \geq 0} \left(n + \frac{1}{2}\right) \frac{f_n(\theta)}{r^{n+\frac{3}{2}}}$$

so, writing the first term (with index $n = 0$) of the second sum separately,

$$\begin{aligned} \frac{\partial}{\partial r} u - iku(r, \theta) &\sim -\frac{1}{2} e^{ikr} \frac{f_0(\theta)}{r^{\frac{3}{2}}} - e^{ikr} \sum_{n \geq 1} \left(n + \frac{1}{2}\right) \frac{f_n(\theta)}{r^{n+\frac{3}{2}}} \\ &= -\frac{1}{2} u(r, \theta) + \mathcal{O}\left(r^{-\frac{5}{2}}\right) - e^{ikr} \sum_{n \geq 1} \left(n + \frac{1}{2}\right) \frac{f_n(\theta)}{r^{n+\frac{3}{2}}} \\ &= -\frac{1}{2r} u(r, \theta) + \mathcal{O}\left(r^{-\frac{5}{2}}\right). \end{aligned}$$

This implies that

$$B_1[u] := \left(\frac{\partial}{\partial r} - ik + \frac{1}{2r}\right) u(r, \theta) = \mathcal{O}\left(r^{-\frac{5}{2}}\right).$$

Note that by using this artificial boundary condition we have that is

- easy to implement for m small
- We do have a systematic error
- Compared to the DtN approach in general this does not couple the variables hence the system-matrix will not be dense but sparse.
- This however, is no exact condition.

We shall now prove the convergence:

Theorem 6.1. *Consider the equation*

$$\begin{cases} \Delta u + k^2 u = 0 & \text{in } \Omega_\infty \subseteq \mathbb{R}^2, \\ \lim_{r \rightarrow +\infty} \sqrt{r}(\partial_r u - iku) = 0, \\ u = g & \text{on } \Gamma := \partial\Omega_\infty. \end{cases}$$

If v satisfies

$$\begin{cases} \Delta v + k^2 v = 0 & \text{in } \Omega_\infty \subseteq \mathbb{R}^2, \\ B_1[v] = 0 & \text{on } B = \partial B_R(0), \\ v = g & \text{on } \Gamma := \partial\Omega_\infty, \end{cases}$$

Then

$$\|u - v\|_{L^2(B)} \leq \frac{C}{kR^2}$$

Proof. The error $w := u - v$ satisfies

$$\begin{cases} \Delta w + k^2 w = 0, & \text{in } \Omega_\infty \subseteq \mathbb{R}^2, \\ B_1[w] = B_1[u], & \text{on } B = \partial B_R(0), \\ w = 0, & \text{on } \Gamma, \end{cases}$$

Then, using integration by parts

$$\begin{aligned} \mathbb{R} \ni \int_{\Omega} |\nabla w|^2 - k^2 |w|^2 dx &= \int_B \bar{w} \partial_r w dS = \int_B \bar{w} \left(ikw - \frac{1}{2R} w + B_1[u] \right) dS \\ &= \|w\|_{L^2(B)}^2 \left(ik - \frac{1}{2R} \right) + \langle B_1[u], w \rangle_{L^2(B)} \end{aligned}$$

The imaginary part must be zero, hence

$$\|w\|_{L^2(B)}^2 = -\frac{1}{k} \operatorname{Im} \langle B_1[u], w \rangle_{L^2(B)} \stackrel{\text{C-S}}{\leq} \frac{1}{k} \|B_1[u]\|_{L^2(B)} \|w\|_{L^2(B)}.$$

Using this and $B_1[u] = \mathcal{O}(R^{-5/2})$ for $R \rightarrow +\infty$ we conclude

$$\|B_1[u]\|_{L^2(B)} = \left[\int_0^{2\pi} |B_1[u](R, \theta)|^2 R d\theta \right]^{\frac{1}{2}} = \mathcal{O}(R^{-2}), \quad R \rightarrow +\infty.$$

□

Question 7. Consider the finite-difference (FD) method applied to the second-order wave equation with leapfrog time-integration. Show (discrete) energy conservation under the resulting CFL stability condition. State the convergence theorem w.r.t. to the energy norm.

Solution 7. The wave equation is given by

$$\begin{cases} \partial_t^2 u - \operatorname{div}(c^2(x) \nabla u) = F(t, x), & \text{in } \Omega \text{ and for } t > 0 \\ u(t, x) = g_D(t, x), & x \in \Gamma_D, t > 0 \\ \partial_n u(t, x) = g_N(t, x), & x \in \Gamma_N, t > 0 \\ u(0, \cdot) = f, & \text{in } \Omega \\ \partial_t u(0, \cdot) = g, & \text{in } \Omega \end{cases}$$

The "continuous" energy is given by

$$E[u](t) = \frac{1}{2} \int_{\Omega} \partial_t u^2 + c(x)^2 |\nabla u|^2 dx$$

and its conservation is proved by taking the derivative w.r.t. to t which also implies the uniqueness of the solution.

We study the 1-D case with $c \equiv 1$, $\Omega = (0, 1)$ and boundary conditions $u(t, 0) = u(t, 1) = 0$. Choosing an equidistant grid in space $x_j = j\Delta x$, $\Delta x = \frac{1}{n+1}$ and time steps $t_m = m\Delta t$. We approximate $v_j^m \approx su(t_m, x_j)$:

$$\frac{v_j^{m+1} - 2v_j^m + v_j^{m-1}}{\Delta t^2} = \frac{v_{j+1}^m - 2v_j^m + v_{j-1}^m}{\Delta x^2}, \quad j = 1, \dots, n, \quad m \geq 1.$$

Set $\mathbf{v}^m := [v_i^m]_{i=1}^n$ and $r = \Delta t / \Delta x$ thus we obtain

$$(7.1) \quad \mathbf{v}^{m+1} = (2\mathbf{I} - r^2 \mathbf{A}) \mathbf{v}^m - \mathbf{v}^{m-1}, \quad \mathbf{A} = \begin{bmatrix} 2 & -1 & & \\ -1 & \ddots & \ddots & \\ & \ddots & \ddots & -1 \\ & & -1 & 2 \end{bmatrix}$$

This method requires two starting methods which we determine by the initial condition and by Taylor expansion:

$$(7.2) \quad v_j^0 = f(x_j), \quad j = 1, \dots, n, \quad \text{and} \quad v_j^1 := v_j^0 + \frac{\Delta t^2}{2} \frac{1}{\Delta x^2} [-\mathbf{A} \mathbf{v}^0]_j, \quad j = 1, \dots, n$$

One needs to choose r such that the matrix $2\mathbf{I} - r^2 \mathbf{A}$ is symmetric and positive definite which is achieved for $\Delta t \leq \Delta x$ or in more generality for $c > 0$ but still constant $\Delta t \leq \Delta x / c$.

Energy conservation. Define the inner product on $\mathbb{R}^n$,

$$\langle \mathbf{v}, \mathbf{w} \rangle_h := \Delta x \cdot \mathbf{v}^\top \mathbf{w}$$

Note that the corresponding norm satisfies for $u : \mathbb{R} \rightarrow \mathbb{R}$ continuous

$$\| [u(x_i)]_{i=1}^n \|_h^2 = \Delta x \sum_{j=1}^n u(x_j)^2 \approx \int_0^1 u^2 dx = \|u\|_{L^2}^2.$$

in particular $\|\mathbf{v}\|_h$ remains bounded as $\Delta x \rightarrow 0$.

Theorem 7.1. Assume that $\{\mathbf{v}^m\}_{m \geq 0}$ is given by a Leapfrog-scheme

$$(7.3) \quad \frac{\mathbf{v}^{m+1} - 2\mathbf{v}^m + \mathbf{v}^{m-1}}{\Delta t^2} = -\frac{1}{\Delta x^2} \mathbf{A} \mathbf{v}^m$$

where $\mathbf{A} \in \mathbb{R}^{n \times n}$ is symmetric and positive definite and let E be the discrete energy given by

$$E[\mathbf{u}, \mathbf{v}] := \frac{1}{2} \left\| \frac{\mathbf{u} - \mathbf{v}}{\Delta t} \right\|_h^2 + \frac{1}{2\Delta x^2} \langle \mathbf{A} \mathbf{u}, \mathbf{v} \rangle_h, \quad \mathbf{u}, \mathbf{v} \in \mathbb{R}^n.$$

Then the quantity

$$E_{m+\frac{1}{2}} := E[\mathbf{v}^m, \mathbf{v}^{m+1}]$$

is constant in m . Also if $\Delta t > 0$ is sufficiently small, then the quadratic form E is positive definite.

Proof. Multiplying (7.3) from the left with $(\mathbf{v}^{m+1} - \mathbf{v}^{m-1})^\top$ yields $E_{m-1/2} = E_{m+1/2}$ for all $m \geq 1$. Also writing

$$\begin{aligned} \frac{E[\mathbf{u}, \mathbf{v}]}{\Delta x} &= \frac{1}{2} \left\| \frac{\mathbf{v} - \mathbf{u}}{\Delta t} \right\|^2 + \frac{1}{2\Delta x^2} \mathbf{v}^\top \mathbf{A} \mathbf{u} \\ &= \frac{1}{2} \left(\frac{\mathbf{v} - \mathbf{u}}{\Delta t} \right)^\top \left[\mathbf{I} - \frac{\Delta t^2}{4\Delta x^2} \mathbf{A} \right] \left(\frac{\mathbf{v} - \mathbf{u}}{\Delta t} \right) + \frac{1}{2\Delta x^2} \left(\frac{\mathbf{v} + \mathbf{u}}{2} \right)^\top \mathbf{A} \left(\frac{\mathbf{v} + \mathbf{u}}{2} \right). \end{aligned}$$

The matrix $\mathbf{I} - \frac{\Delta t^2}{4\Delta x^2} \mathbf{A}$ is symmetric for $\Delta t > 0$ sufficiently small also positive definite which implies positive semi-definiteness. If $E[\mathbf{u}, \mathbf{v}] = 0$ we must have

$$\mathbf{v} - \mathbf{u} = \mathbf{0} \quad \text{and} \quad \mathbf{v} + \mathbf{u} = \mathbf{0}$$

and thus $\mathbf{u} = \mathbf{v} = \mathbf{0}$. This implies positive definiteness.

Theorem 7.2. Let $u \in C^4([0, T] \times [0, 1])$ be the solution of the problem

$$\begin{cases} \partial_t^2 u - \partial_x^2 u = F(t, x), & (t, x) \in [0, T] \times (0, 1), \\ u(t, 0) = u(t, 1) = 0, & t \in (0, T), \\ u(0, \cdot) = f, \\ \partial_t u(0, \cdot) = g \end{cases}$$

Suppose u is approximated by the numerical solution $v_j^m \approx u(t_m, x_j)$ obtained by (7.1) $r = \frac{\Delta t}{\Delta x} < 1$ and such that v_j^0 and v_j^1 satisfy (7.2). Then the error $e_j^m = u(t_m, x_j) - v_j^m$ satisfies

$$\sqrt{E_{m+\frac{1}{2}}} := \sqrt{E[e^{m+1}, e^m]} = \mathcal{O}(\Delta x^2), \quad t_{m+1} \in (0, T], \Delta x \rightarrow 0. \quad \square$$

Question 8. Derive the semi-discrete and fully discrete Galerkin FE formulation of the second-order wave equation with leapfrog time integration. Explain mass-lumping in 1-D and in 2-D for triangular and quadrilateral FE.

Solution 8. Consider

$$\begin{cases} \partial_t^2 u - \operatorname{div}(c(x)^2 \nabla u) = f, & \text{in } (0, T) \times \Omega, \\ u = 0, & \text{on } (0, T) \times \Gamma, \\ u(0, \cdot) = u_0, & \text{in } \Omega, \\ \partial_t u(0, \cdot) = v_0, & \text{in } \Omega, \end{cases}$$

We first need the variational formulation: Find $u : [0, T] \rightarrow V =: H_0^1(\Omega)$ such that

$$(8.1) \quad \frac{d^2}{dt^2} \langle u(t), v \rangle_{L^2} + a(u(t), v) = \langle f(t), v \rangle_{L^2}, \quad u(0) = u_0, \quad \left. \frac{d}{dt} u(t) \right|_{t=0} = v_0.$$

Semi-discrete Galerkin scheme. We approximate $V = H_0^1(\Omega)$ by a discrete subspace V_h . The semi-discrete formulation then is: find $u_h \in \mathcal{C}^2((0, T); V_h)$ such that

$$\begin{aligned} \frac{d^2}{dt^2} \langle u_h(t), v \rangle_{L^2} + a(u_h(t), v) &= \langle f(t), v \rangle, \quad \text{for all } v \in V_h, t \in (0, T), \\ u_h(0) &= I_h u_0, \quad \left. \frac{d}{dt} u_h(t) \right|_{t=0} = I_h v_0. \end{aligned}$$

Then take a basis $\varphi_1, \dots, \varphi_N$ of V_h and write everything in coordinates w.r.t. that basis. i.e., one gets

$$\mathbf{M} \frac{d^2}{dt^2} \mathbf{u}(t) + \mathbf{A} \mathbf{u}(t) = \mathbf{F}(t)$$

where

$$\underbrace{\mathbf{M} = [\langle \varphi_i, \varphi_j \rangle]_{i,j=1}^N}_{\text{mass matrix}}, \quad \underbrace{\mathbf{A} = [a(\varphi_i, \varphi_j)]_{i,j=1}^N}_{\text{stiffness matrix}}, \quad \mathbf{F}(t) = [\langle f(t), \varphi_i \rangle_{L^2}]_{i=1}^N$$

Note that the semi-discrete problem still conserves the energy

$$E[u_h](t) = \frac{1}{2} \left(\|u_h'(t)\|^2 + a(u_h, u_h) \right)$$

Fully discrete Galerkin approximation (Leapfrog). Introduce discretization in time $t_m = m\Delta t$, $\Delta t > 0$ and let $\mathbf{u}^m \approx \mathbf{u}(t_m)$, $\mathbf{F}^m = \mathbf{F}(t_m)$ and replace the second order derivatives centered finite differences giving

$$\boxed{\mathbf{M} \mathbf{u}^{m+1} = (2\mathbf{M} - \Delta t^2 \mathbf{A}) \mathbf{u}^m - \mathbf{M} \mathbf{u}^{m-1} + \Delta t^2 \mathbf{F}^m.}$$

The "problem" here is that we need to solve a linear system in any step. The solution here is "mass lumping"

Mass lumping. Note that the matrix $\mathbf{M}$ is easily invertible if it is diagonal. This can be achieved using a quadrature formula. Working in 1-D, instead of

$$M_{i,j} = \langle \varphi_i, \varphi_j \rangle_{L^2} = \sum_{k=0}^N \int_{x_k}^{x_{k+1}} \varphi_i \varphi_j \, dx$$

one replaces the integration by the trapezoidal rule:

$$\int_{x_k}^{x_{k+1}} f(x) \, dx \approx \frac{h}{2} (f(x_k) + f(x_{k+1})).$$

which gives using $\varphi_i(x_k) = \delta_{i,k}$, and the fact that trapezoidal rule is exact for $\mathcal{P}^1$ -elements that

$$\tilde{M}_{i,j} = \delta_{i,j} \sum_{k=0}^N \frac{h}{2} \left(\varphi_i(x_k) + \varphi_i(x_{k+1}) \right) = \delta_{i,j} \sum_{k=0}^N \int_{x_k}^{x_{k+1}} \varphi_i \, dx = \delta_{i,j} \int_{\Omega} \varphi_i(x) \, dx$$

that is

$$\int_{\Omega} \varphi_i \, dx = \int_{\Omega} 1 \cdot \varphi_i(x) \, dx = \boldsymbol{\varphi}_i^T \mathbf{M} \mathbf{1} = \sum_{k=0}^{N+1} M_{i,k},$$

i.e.,

$$\tilde{M}_{i,j} = \delta_{i,j} \sum_{k=0}^{N+1} M_{i,k}.$$

If we are working in 2-D we can use the quadrature rule on the reference element $\hat{K}$,

$$Q_{\hat{K}}[f] = \underbrace{|\hat{K}|}_{=1/2} \frac{1}{3} \sum_{i=0}^2 f(\boldsymbol{\xi}^{(i)}) \approx \int_{\hat{K}} f(\boldsymbol{\xi}) \, d\boldsymbol{\xi},$$

where $\{\boldsymbol{\xi}^i\}_{i=0}^2$ are the mid-points on each edge of the reference triangle $\hat{K}$. This quadrature rule is exact for every $f \in \mathcal{P}^1(\hat{K})$. Hence we compute the mass-matrix via

$$M_{i,j} = \sum_{\substack{K \in \mathcal{T} \\ x_p, x_q \in K}} |\det \mathbf{J}_K| \underbrace{\int_{\hat{K}} N_p(\boldsymbol{\xi}) N_q(\boldsymbol{\xi}) d\boldsymbol{\xi}}_{=: \hat{M}_{p,q}},$$

where $p, q = 0, 1, 2$ are the local indices in K corresponding to the (global numbering) j, i . One can show that

$$\hat{\mathbf{M}} = \frac{1}{24} \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}.$$

Replacing the exact integration by this quadrature formula and using

$$N_q(\boldsymbol{\xi}^{(k)}) N_p(\boldsymbol{\xi}^{(k)}) = \delta_{p,q} N_p(\boldsymbol{\xi}^{(k)})$$

one obtains

$$[\tilde{\mathbf{M}}_{\hat{K}}]_{p,q} = \frac{1}{6} \sum_{k=0}^2 N_q(\boldsymbol{\xi}^{(k)}) N_p(\boldsymbol{\xi}^{(k)}) = \frac{\delta_{p,q}}{6} \sum_{k=0}^2 N_p(\boldsymbol{\xi}^{(k)}) = \frac{\delta_{p,q}}{6},$$

that is, $\tilde{\mathbf{M}}_{\hat{K}} = \frac{1}{6} \cdot \mathbf{I}$ and thus

$$\tilde{M}_{i,j} = \delta_{i,j} \sum_{k \in \mathcal{N}} M_{i,k}$$

analog to the 3-D case. For higher order finite elements one can extend this idea. However one needs to choose the right quadrature rule. This requires sacrificing a bit of the exactness of the quadrature formula (c.f. Gauss-Lobato elements). $\square$

Before we discuss the next questions we need to make some comments about von Neumann-stability analysis.

NOTES ON VON NEUMANN-STABILITY ANALYSIS

The Von Neumann stability analysis (see also next question) is a method to prove stability and convergence of finite difference methods on uniform equidistant meshes. We will consider the equation

$$(8.2) \quad \begin{cases} \partial_t u + c \partial_x u = 0, & \text{in } (0, T) \times \mathbb{R} \\ u(0, \cdot) = f, & x \in \mathbb{R} \end{cases}$$

with c constant and f 1-periodic and look at one-step explicit schemes of the form

$$(8.3) \quad v_j^{m+1} = \sum_{\ell=\alpha}^{\beta} Q_{\ell} v_{j+\ell}^m, \quad m \geq 0.$$

where the mesh is given by $x_j = j\Delta x$, $\Delta x = \frac{1}{n+1}$, $j \in \mathbb{N}$ and $t_m = m\Delta t$ with $\Delta t > 0$. Also $r := \Delta t / \Delta x$ will always be constant. A function with domain $\{x_j\}_{j \in \mathbb{N}}$ is called *grid function* and is 1-periodic if $v(x_j) =: v_j = v_{j+n+1}$. Define further $\langle v, w \rangle_h = \Delta x \sum_{j=0}^n v_j \bar{w}_j$.

Definition. Fix $r > 0$.

(1) Let u be a smooth solution of (8.2). The *local truncation error* τ is given by

$$(8.4) \quad \Delta t \tau_j^m = u(x_j, t_{m+1}) - \sum_{\ell=\alpha}^{\beta} Q_{\ell} u(x_{j+\ell}, t_m).$$

The finite different scheme (8.3) is accurate of order p , if for every time interval length T and sufficiently smooth solution u of (8.2) there exists a constant C_T such that

$$\|\tau^m\|_h \leq C_T \Delta x^p, \quad \text{for all } t_m \in [0, T].$$

for every Δx sufficiently small (recall $r = \frac{\Delta t}{\Delta x}$ is fixed). If $p > 0$ we say that the scheme is consistent.

- (2) The FD scheme is stable for $r > 0$ if for any $T > 0$ there exists a constant C_T such that for every Δx sufficiently small and for any 1-periodic grid function v^0 ,

$$\|v^m\|_h \leq C_T \|v^0\|_h \quad \text{for any } t_m \in [0, T].$$

Theorem 8.1. *Suppose the Cauchy problem (8.2) is well posed and the FD scheme (8.3) is consistent (or accurate of order p). Then the FD scheme (8.3) is convergent (of order p) if, and only if, it is stable.*

Proof. Suppose the scheme is stable and assume for simplicity that the scheme is given by

$$\mathbf{v}^{m+1} = \mathbf{Q}\mathbf{v}^m$$

(works analogous for more complicated schemes). Then the error $e_j^m = u(x_j, t_m) - v_j^m$ satisfies

$$\mathbf{e}^{m+1} \stackrel{(8.4)}{=} \mathbf{Q}\mathbf{e}^m + \Delta t \boldsymbol{\tau}^m$$

and thus,

$$\begin{aligned} \|\mathbf{e}^m\|_h &= \left\| \mathbf{Q}^m \mathbf{e}^0 + \Delta t \sum_{k=0}^{m-1} \mathbf{Q}^{m-1-k} \boldsymbol{\tau}^k \right\|_h \\ &\leq C_T \|\mathbf{e}^0\|_h + C_T T \max_{k=0}^{m-1} \|\boldsymbol{\tau}^k\|_h \\ &\leq C_t \|\mathbf{e}^0\|_h + \tilde{C}_T T \Delta x^p \end{aligned}$$

where we used that $T = m \cdot \Delta t$ and that the scheme is stable and in the last step that it is consistent.

Given a grid function $v_j = v(x_j)$ that is 1-periodic we define its *Fourier-transform* by

$$\tilde{v}_k = \Delta x \sum_{j=0}^n v_j e^{-2\pi i k x_j} = \Delta x \sum_{j=0}^n v_j e^{-i \eta k j}, \quad \eta = 2\pi \Delta x$$

We have the following properties of this Fourier transform:

Lemma 8.1. (1) *The map $v \mapsto \tilde{v}$ is invertible with*

$$v_\ell = \sum_{k=0}^n \tilde{v}_k e^{i \eta k \ell}$$

(2) *We have a Parseval-type equality:*

$$\langle v, w \rangle_h = \sum_{k=0}^n \overline{\tilde{w}_k} \tilde{v}_k, \quad \text{and} \quad \|v\|_h = \sum_{k=0}^n |\tilde{v}_k|^2.$$

(3) *If for $\ell \in \{0, \dots, n\}$ fixed, if $w_j = v_{j+\ell}$, then*

$$\tilde{w}_k = \tilde{v}_k e^{i \ell \eta k} \quad \text{for all } k.$$

We now can prove the following:

Theorem 8.2. *The scheme (8.3) is stable for $r > 0$ if and only if there exist constants n_0, C_T such that for every $n \geq n_0$,*

$$|\hat{Q}(\eta k)^m| \leq C_T, \quad k = 0, \dots, n, \quad t_m = m \Delta t \in [0, T]$$

where $\Delta x = \frac{1}{n+1}$, $\eta = 2\pi \Delta x$ and $\Delta t = r \Delta x$.

Proof. Suppose that the scheme (8.3) is stable. By Parseval we have

$$\|v^m\|_h = \sum_{k=0}^n |\tilde{v}_k^m|^2 = \sum_{k=0}^n |\hat{Q}(\eta k)^m \tilde{v}_k^0|^2.$$

Fix $\ell = 0, \dots, n$ and then choose v^0 such that $\tilde{v}_k^0 = 0$ for $k \neq \ell$ and $\tilde{v}_\ell^0 = \xi$. Then we infer that

$$\begin{aligned} |\hat{Q}(\eta \ell)^m|^2 &= \sum_{k=0}^n |\hat{Q}(\eta k)^m \tilde{v}_k^0|^2 \\ &= \|v^m\|_h^2 \stackrel{\text{Stability}}{\leq} C_T^2 \|v^0\|_h^2 \\ &= C_T^2 \sum_{k=0}^n |\tilde{v}_k^0|^2 = C_T^2. \end{aligned}$$

which holds for all ℓ as the RHS is independent of ℓ . The other implication is rather straightforward.

Question 9. Consider the first-order hyperbolic Cauchy problem

$$(9.1) \quad \begin{cases} \partial_t u(t, x) + c(x) \partial_x u(t, x) = F(t, x), & x \in \mathbb{R}, t > 0 \\ u(0, x) = f(x), & x \in \mathbb{R}, \end{cases}$$

with c and f 1-periodic. Show that the Lax-Friedrichs method (on an equidistant mesh) converges as $\mathcal{O}(\Delta x) + \mathcal{O}(\Delta t)$ with respect to $\|\cdot\|_{L^\infty}$.

Solution 9. The Lax-Friedrichs scheme is given by

$$(9.1) \quad v_j^{m+1} = \frac{1}{2} (v_{j+1}^m + v_{j-1}^m) - \frac{c_j \Delta t}{2 \Delta x} (v_{j+1}^m - v_{j-1}^m).$$

Then for u smooth enough one has

$$\begin{aligned} \Delta t \tau_j^m &= u_j^{m+1} - \left[\frac{1}{2} (u_{j+1}^m + u_{j-1}^m) - \frac{c_j \Delta t}{2 \Delta x} (u_{j+1}^m - u_{j-1}^m) \right] \\ &= u_j^{m+1} - \left[\frac{1}{2} (1 - c_j r) u_{j+1}^m + \frac{1}{2} (1 + c_j r) u_{j-1}^m \right] \end{aligned}$$

where $u_j^m = u(x_j, t_m)$. Using Taylor-expansion in around (x_j, t_m) in every coordinate one gets

$$\begin{aligned} \tau_j^m &= \underbrace{u(x_j, t_m) + c_j u_x(x_j, t_m)}_{\text{PDE} \Rightarrow 0} + \frac{\Delta t}{2} u_{tt}(x_j, s_m) \\ &\quad - \frac{\Delta x}{4r} \left((1 - c_j r) u_{xx}(\xi_j, t_m) + (1 + c_j r) u_{xx}(\tilde{\xi}_j, t_m) \right) \end{aligned}$$

for some $s_m \in (t_m, t_{m+1})$, $\xi_j \in (x_j, x_{j+1})$, $\tilde{\xi}_j \in (x_{j-1}, x_j)$. Thus, if $u \in \mathcal{C}^2$, then

$$|\tau_j^m| \leq \tilde{C} \max_{(t,x) \in [0,T] \times [0,1]} \max \left\{ |\partial_t^2 u(t, x)|, |\partial_x^2 u(t, x)| \right\} (\Delta x + \Delta t),$$

where $\tilde{C}$ only depends on $c_{\max}$. This means that the scheme is consistent.

To prove convergence one takes $c_{\max} \frac{\Delta t}{\Delta x} \leq 1$ and computes for $e_j^m := u(t_m, x_j) - v_j^m$, ($r := \Delta t / \Delta x$)

$$\begin{aligned} |e_j^{m+1}| &= \left| \frac{1}{2} \underbrace{(1 - c_j r)}_{\geq 0} e_{j+1}^m + \frac{1}{2} \underbrace{(1 + c_j r)}_{\geq 0} e_{j-1}^m + \Delta t \tau_j^m \right| \\ &\leq \|e^m\|_\infty + \Delta t \|\tau^m\|_\infty \end{aligned}$$

and thus

$$\|e^m\|_\infty \leq \|e^0\|_\infty + \Delta t \sum_{k=0}^{m-1} \|\tau^k\|_\infty \leq t_m \max_{k=0}^{m-1} \|\tau^k\|_\infty = \mathcal{O}(\Delta t + \Delta x).$$

□

Question 10. Consider the previous first-order hyperbolic Cauchy problem (9.1) with $c > 0$ and constant. Show stability and convergence of a first-order up winding FD method on an equidistant mesh using Von Neumann stability analysis and the Lax equivalence theorem.

Solution 10. Before answering the question we discuss some basic facts about the upwinding method.

Remark. Upwinding methods are one-sided finite difference methods. Considering (9.1) we get, approximating the derivative from the right and the left, the FD schemes

$$(10.1) \quad \frac{v_j^{m+1} - v_j^m}{\Delta t} + c_j \frac{v_j^m - v_{j-1}^m}{\Delta x} = 0$$

$$(10.2) \quad \frac{v_j^{m+1} - v_j^m}{\Delta t} + c_j \frac{v_{j+1}^m - v_j^m}{\Delta x} = 0$$

The reason for using less accurate finite differences is that they preserve the characteristic curves. For $c(x) > 0$, say, the exact solution of (9.1) given by $u(X(t; x_0), t) = f(x_0)$ transports the initial condition along characteristic curves which point from left to right only. Hence for $c > 0$, information in the PDE propagates only from left to right. The upwind scheme preserves this property. In particular, for constant $c > 0$ and $c \frac{\Delta t}{\Delta x} = 1$ we have

$$v_j^{m+1} = v_j^m - c \frac{\Delta t}{\Delta x} (v_j^m - v_{j-1}^m) = v_{j-1}^m.$$

Since

$$u(x_j, t_{m+1}) = f(x_j - ct_{m+1}) = f(x_j - ct_m + c\Delta t) = f(x_j - \Delta x - ct_m) = u(x_{j-1}, t_m)$$

that is, the finite difference approximation (10.1) in this special case is exact. For $c(x) > 0$, (10.1) is called an upwind scheme, and, similarly, for $c(x) < 0$, (10.2) is an upwind scheme. Similarly, for $c < 0$ and constant, the FD scheme (10.2) with $-1 = c \frac{\Delta t}{\Delta x}$ is exact.

When looking at $0 < c(x) \leq c_{\max}$ one determines using the Domain of dependence of the discrete and continuous solution that the CFL condition necessary for convergence is

$$\boxed{c_{\max} \frac{\Delta t}{\Delta x} \leq 1.}$$

Note that the FD scheme (10.2) cannot converge for $c > 0$, as the domain of dependence of the numerical solution never contains that of the exact solution, regardless of Δt . [c.f. notes, page 79]

We apply the discrete Fourier transform to (10.1) with $c \neq 0$ constant by formally substituting $v_j^m = \tilde{v}_k^m e^{i\eta k j}$ where $\eta = 2\pi \Delta x$. Setting $c \frac{\Delta t}{\Delta x} =: \nu$ we get

$$\tilde{v}_k^m = a_k^m \tilde{v}_k^0, \quad \text{where} \quad 1 - \nu (1 - e^{i\eta k}).$$

Using

$$|a_k| = 1 + 2\nu(\nu - 1)(1 - \cos(k\eta)) \stackrel{!}{\leq} 1$$

this leads to $\nu \in [0, 1]$. For $c < 0$ the scheme is unconditionally unstable. One can also prove that the error e_m^j of the upwind method

$$e_j^m = u(x_j, t_m) - v_j^m \quad \text{satisfies} \quad \max_{k=1, \dots, m} \|e^k\| = \mathcal{O}(\Delta x, \Delta t).$$

Theorem 10.1 (Theorem IV.2.10). Let $u \in C^2([0, 1] \times [0, T])$ be the solution of (P) with $0 < c(x) \leq c_{\max}$, and let $v_j^m \approx u(x_j, t_m)$ be the discrete solution with the upwind method

$$\frac{v_j^{m+1} - v_j^m}{\Delta t} + c_j \frac{v_j^m - v_{j-1}^m}{\Delta x} = 0 \quad \implies \quad v_j^{m+1} = (1 - c_j r) v_j^m + c_j r v_{j-1}^m, \quad r = \frac{\Delta t}{\Delta x}$$

with $v_j^0 = f(x_j)$. If $c_{\max} \frac{\Delta t}{\Delta x} \leq 1$, then the error $e_j^m := u(x_j, t_m) - v_j^m$ satisfies

$$\max_{1 \leq k \leq m} \|e^k\|_\infty \leq Ct_m(\Delta t + \Delta x) = O(\Delta x + \Delta t).$$

Proof. Step 1: Local Truncation Error Analysis

Let $u_j^m := u(x_j, t_m)$ denote the exact solution evaluated on the grid. By definition, the local truncation error τ_j^m for the upwind scheme is given by

$$\Delta t \tau_j^m = u_j^{m+1} - [(1 - c_j r)u_j^m + c_j r u_{j-1}^m] = u_j^{m+1} - u_j^m + c_j \frac{\Delta t}{\Delta x} (u_j^m - u_{j-1}^m).$$

Using Taylor expansions of $u \in C^2$ around (x_j, t_m) :

$$u_j^{m+1} = u_j^m + \Delta t u_t(x_j, t_m) + \frac{\Delta t^2}{2} u_{tt}(x_j, s_m), \quad s_m \in (t_m, t_{m+1})$$

$$u_{j-1}^m = u_j^m - \Delta x u_x(x_j, t_m) + \frac{\Delta x^2}{2} u_{xx}(\xi_j, t_m), \quad \xi_j \in (x_{j-1}, x_j)$$

Substituting these Taylor expansions into the expression for τ_j^m :

$$\begin{aligned} \tau_j^m &= \frac{u_j^{m+1} - u_j^m}{\Delta t} + c_j \frac{u_j^m - u_{j-1}^m}{\Delta x} \\ &= (u_t(x_j, t_m) + c(x_j)u_x(x_j, t_m)) + \frac{\Delta t}{2} u_{tt}(x_j, s_m) - \frac{c_j \Delta x}{2} u_{xx}(\xi_j, t_m). \end{aligned}$$

Since u satisfies the PDE $u_t + c(x)u_x = 0$, the first term vanishes:

$$\tau_j^m = \frac{\Delta t}{2} u_{tt}(x_j, s_m) - \frac{c(x_j) \Delta x}{2} u_{xx}(\xi_j, t_m).$$

Taking the maximum norm over all spatial grid points yields:

$$\|\tau^m\|_\infty \leq \frac{\Delta t}{2} \|u_{tt}\|_\infty + \frac{c_{\max} \Delta x}{2} \|u_{xx}\|_\infty \leq C_0(\Delta t + \Delta x)$$

where $C_0 = \frac{1}{2} \max(\|u_{tt}\|_\infty, c_{\max} \|u_{xx}\|_\infty) < \infty$.

Step 2: Error Recursion Equation

Subtracting the numerical scheme from the truncation error relation yields the error equation for $e_j^m = u_j^m - v_j^m$:

$$e_j^{m+1} = (1 - c_j r) e_j^m + c_j r e_{j-1}^m + \Delta t \tau_j^m.$$

Under the CFL condition $c_{\max} r = c_{\max} \frac{\Delta t}{\Delta x} \leq 1$ and $c(x) > 0$, the coefficients satisfy:

$$1 - c_j r \geq 0 \quad \text{and} \quad c_j r \geq 0 \quad \forall j.$$

Step 3: Maximum Norm Estimate

Taking absolute values on both sides of the error recursion and applying the triangle inequality:

$$\begin{aligned} |e_j^{m+1}| &\leq (1 - c_j r) |e_j^m| + c_j r |e_{j-1}^m| + \Delta t |\tau_j^m| \\ &\leq (1 - c_j r) \|e^m\|_\infty + c_j r \|e^m\|_\infty + \Delta t \|\tau^m\|_\infty \\ &= (1 - c_j r + c_j r) \|e^m\|_\infty + \Delta t \|\tau^m\|_\infty = \|e^m\|_\infty + \Delta t \|\tau^m\|_\infty. \end{aligned}$$

Taking the maximum over all j gives:

$$\|e^{m+1}\|_\infty \leq \|e^m\|_\infty + \Delta t \|\tau^m\|_\infty.$$

Step 4: Summation over Time Steps

Unrolling the recursion down to $m = 0$:

$$\|e^m\|_\infty \leq \|e^0\|_\infty + \Delta t \sum_{k=0}^{m-1} \|\tau^k\|_\infty.$$

Since the initial condition is exact ($v_j^0 = f(x_j) = u(x_j, 0)$), $\|e^0\|_\infty = 0$. Using the uniform bound $\|\tau^k\|_\infty \leq C_0(\Delta t + \Delta x)$:

$$\Delta t \sum_{k=0}^{m-1} \|\tau^k\|_\infty \leq \Delta t \cdot m \cdot \max_{0 \leq k \leq m-1} \|\tau^k\|_\infty = t_m \max_{0 \leq k \leq m-1} \|\tau^k\|_\infty.$$

Therefore:

$$\|e^m\|_\infty \leq C_0 t_m (\Delta t + \Delta x).$$

Taking the maximum over all time levels $1 \leq k \leq m$:

$$\max_{1 \leq k \leq m} \|e^k\|_\infty \leq C t_m (\Delta t + \Delta x) = O(\Delta t + \Delta x)$$

where $C = C_0 = \frac{1}{2} \max(\|u_{tt}\|_\infty, c_{\max}\|u_{xx}\|_\infty)$.

□

Question 11. Consider the semi-discrete Galerkin FE formulation of the second-order wave equation. Show energy conservation. State the convergence theorem w.r.t. to the energy norm and give the main steps of the proof.

Solution 11. The energy conservation of the wave equation is shown by taking the derivative of

$$E[u](t) = \frac{1}{2} \left(\|u'(t)\|_{L^2}^2 + a(u(t), u(t)) \right)$$

and showing that it is zero. Also, in the discrete case one shows analogously that

$$E[u_h](t) = \frac{1}{2} \left((\mathbf{u}')^\top \mathbf{M} \mathbf{u}' + \mathbf{u}^\top \mathbf{A} \mathbf{u} \right)$$

is constant.

The semi discrete Galerkin formulation is equivalent to (see Question 8)

$$\mathbf{M} \frac{d^2}{dt^2} \mathbf{u}(t) + \mathbf{A} \mathbf{u}(t) = \mathbf{F}(t).$$

We have the following theorem.

Theorem 11.1. Consider the variational formulation

$$(11.1) \quad \begin{cases} \text{Find } u : [0, T] \rightarrow V = H_0^1(\Omega) \text{ such that} \\ \frac{d^2}{dt^2}(u(t), v) + a(u(t), v) = (f(\cdot, t), v), \quad \forall v \in V, \\ u(0) = u_0, \quad \frac{d}{dt}u(0) = v_0, \text{ where} \\ a(u, v) = \int_{\Omega} c^2(x) \nabla u \cdot \nabla v \, dx. \end{cases}$$

Suppose $0 < c_{\min} \leq c(\mathbf{x}) \leq c_{\max}$, for every $\mathbf{x} \in \Omega$ and u_0, v_0 and f are sufficiently differentiable so

$$u \in C^2(0, T; H_0^1(\Omega) \cap H^2(\Omega)).$$

Furthermore, let $u_h \in C^2((0, T); V_h)$ be the solution of the semi-discrete Galerkin approximation of (V) i.e., let u_h satisfy (V) with the space V replaced by

$$V_h = \{v \in C^0(\bar{\Omega}) \mid v|_{\partial\Omega} = 0, v|_K \in \mathcal{P}^1(K) \forall K \in \mathcal{T}_h\} \subseteq V$$

and the initial data u_0, v_0 replaced by $I_h u_0, I_h v_0$. If the triangulation $\{\mathcal{T}_h\}_{h \in \mathcal{H}}$ is a shape regular family of triangulations of Ω then for every sufficiently small $h > 0$,

$$\|(u - u_h)'\|_{C^0((0, T); L^2(\Omega))} + \|u - u_h\|_{C^0((0, T); H^1(\Omega))} \leq Ch \sqrt{T^2 + 1} \|u\|_{C^2((0, T); H^2(\Omega))},$$

where C depends only on $c_{\min}, c_{\max}$ and Ω (and not on u).

Proof. Define the error $e(t) = u(t) - u_h(t)$. Deriving and using an analogous term to Galerkin orthogonality one can show

$$\frac{d}{dt} \left\{ \|e'(t)\|_{L^2}^2 + a(e(t), e(t)) \right\} = \langle e'(t), \rho'(t) \rangle_{L^2} + a(e(t), \rho'(t)),$$

where $\rho(t) = (u - I_h u)(t)$. Integrating this with regard to t over $[0, s]$, where $s \in [0, T]$ one deduces, using integration by parts,

$$\frac{1}{2} \left[\|e'(t)\|_{L^2}^2 + a(e(t), e(t)) \right]_{t=0}^s = \left[\langle e'(t), \rho'(t) \rangle \right]_{t=0}^s - \int_0^s \langle e'(s), \rho''(s) \rangle_{L^2} dt + \int_0^s a(e(t), \rho'(t)) dt.$$

Then use the continuity and coercivity of a to obtain an upper bound independent of s . Also by

$$\sup_s a(s) + \sup_s b(s) \leq 2 \sup_s (a(s) + b(s))$$

we get (using the notation $\mathcal{C}_t^0 L_x^2$ to denote $\mathcal{C}^0((0, T); L^2(\Omega))$ and so on)

$$\begin{aligned} & \|e\|_{\mathcal{C}_t^0 L_x^2}^2 + \alpha \|e\|_{\mathcal{C}_t^0 L_x^2}^2 \\ & \leq 2\|e'(0)\|_{L^2}^2 + 2C\|e(0)\|_{H^1}^2 + 4 \left[2\|\rho'\|_{\mathcal{C}_t^0 L_x^2} + T\|\rho''\|_{\mathcal{C}_t^0 L_x^2} \right] + 4CT\|\rho'\|_{\mathcal{C}_t^0 H_x^1} \|e\|_{\mathcal{C}_t^0 H_x^1}, \end{aligned}$$

where α is the coercivity constant of a and C its continuity constant. Then by the inequalities

$$ab \leq \frac{\varepsilon}{2} a^2 + \frac{1}{2\varepsilon} b^2, \quad \text{and} \quad (a+b)^2 \leq 2(a^2 + b^2) \quad a, b \in \mathbb{R}, \varepsilon > 0,$$

we get upper estimates. Subtracting the terms involving e finally leads to

$$\begin{aligned} & \frac{1}{2} \|e'\|_{\mathcal{C}_t^0 L_x^2}^2 + \frac{\alpha}{2} s \|e\|_{\mathcal{C}_t^0 H_x^1}^2 \\ & \leq 2\|e'(0)\|_{L^2}^2 + 2C\|e(0)\|_{H^1}^2 + 64\|\rho'\|_{\mathcal{C}_t^0 L_x^2}^2 + 16T^2\|\rho''\|_{\mathcal{C}_t^0 L_x^2}^2 + 8\frac{C^2 T^2}{\alpha} \|\rho'\|_{\mathcal{C}_t^0 H_x^1}^2 \end{aligned}$$

At last use that

$$\begin{aligned} \|\rho'\|_{\mathcal{C}_t^0 L_x^2} &= \|(u - I_h u)'\|_{\mathcal{C}_t^0 L_x^2} \lesssim h^2 \|u\|_{\mathcal{C}_t^1 H_x^2} \\ \|\rho''\|_{\mathcal{C}_t^0 L_x^2} &= \|(u - I_h u)''\|_{\mathcal{C}_t^0 L_x^2} \lesssim h^2 \|u\|_{\mathcal{C}_t^2 H_x^2} \\ \|\rho\|_{\mathcal{C}_t^0 L_x^2} &= \|(u - I_h u)\|_{\mathcal{C}_t^0 H_x^1} \lesssim h \|u\|_{\mathcal{C}_t^1 H_x^2} \end{aligned}$$

and

$$\begin{aligned} \|e'(0)\|_{L^2} &= \|u'(0) - I_h u'(0)\|_{L^2} \lesssim h^2 \|u'(0)\|_{H^2} \lesssim h^2 \|u\|_{\mathcal{C}_t^1 H_x^2} \\ \|e(0)\|_{H^1} &= \|u(0) - I_h u(0)\|_{H^1} \lesssim h \|u(0)\|_{H^2} \lesssim \|u\|_{\mathcal{C}_t^0 H_x^2}. \end{aligned}$$

This concludes the proof. □

Question 12. Consider the one-dimensional Cauchy problem for the second-order wave equation

$$\begin{cases} \partial_{t,t} u(t, x) - \partial_{x,x} u(t, x) = F(t, x), & x \in \mathbb{R}, t > 0 \\ u(0, \cdot) = f, & \text{in } \mathbb{R} \\ \partial_t u(0, \cdot) = g, & \text{in } \mathbb{R}. \end{cases}$$

Assuming that $\text{Spt}(f, g, F(t, \cdot)) \subseteq \Omega$ with $\Omega = (0, 1)$, i.e., f , g and F vanish outside Ω , derive the non reflecting boundary conditions at $x = 0, 1$.

Solution 12. Writing $w = \partial_t u$ and $v = \partial_x u$ we have that

$$\frac{\partial}{\partial t} \begin{bmatrix} v \\ w \end{bmatrix} + \underbrace{\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}}_{=: \mathbf{A}} \frac{\partial}{\partial x} \begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} F \\ 0 \end{bmatrix}, \quad \begin{bmatrix} w \\ u \end{bmatrix} (0, \cdot) = \begin{bmatrix} g \\ \partial_x f \end{bmatrix}$$

which suggests studying the equation

$$\begin{cases} \frac{\partial}{\partial t} \mathbf{u} + \mathbf{A} \frac{\partial}{\partial x} \mathbf{u} = \mathbf{f}, & x \in \mathbb{R}, t > 0 \\ \mathbf{u}(0, \cdot) = \mathbf{u}_0, & \text{on } \mathbb{R} \end{cases}$$

Suppose that $\mathbf{A} \in \mathbb{R}^{m \times m}$ is diagonalizable with

$$\mathbf{S}^{-1}\mathbf{A}\mathbf{S} = \begin{bmatrix} \mathbf{\Lambda}^I & & \\ & \mathbf{\Lambda}^{II} & \\ & & \mathbf{\Lambda}^{III} \end{bmatrix} =: \mathbf{\Lambda}$$

where $\mathbf{\Lambda}^I \in \mathbb{R}^{r \times r}$ contains all positive eigenvalues, $\mathbf{\Lambda}^{II} \in \mathbb{R}^{s \times s}$ all negative ones and $\mathbf{\Lambda}^{III} = \mathbf{0}$. Then we get, setting $\mathbf{v} = \mathbf{S}^{-1}\mathbf{u}$ one obtains

$$\partial_t \mathbf{v} + \mathbf{\Lambda} \partial_x \mathbf{v} = \mathbf{S}^{-1} \mathbf{f} =: \tilde{\mathbf{f}}$$

This means we have obtained a system of decoupled equations, where each equation is of the form

$$\partial_t v \pm c \partial_x v = f.$$

□

ANSWER OF CHATGPT TO QUESTION 12

Consider the one-dimensional wave equation

$$\partial_{tt}u - \partial_{xx}u = F(t, x), \quad x \in \mathbb{R}, t > 0,$$

with initial conditions

$$u(0, x) = f(x), \quad \partial_t u(0, x) = g(x),$$

and assume that

$$\text{supp}(f), \text{supp}(g), \text{supp}(F(t, \cdot)) \subset (0, 1).$$

Since the source term vanishes outside the computational domain,

$$F(t, x) = 0, \quad x < 0 \quad \text{or} \quad x > 1,$$

the wave equation reduces to the homogeneous equation

$$\partial_{tt}u - \partial_{xx}u = 0.$$

The general solution of the homogeneous wave equation is given by d'Alembert's formula,

$$u(t, x) = \phi(x - t) + \psi(x + t),$$

where

- $\phi(x - t)$ represents a wave traveling to the right,
- $\psi(x + t)$ represents a wave traveling to the left.

Transparent boundary condition at left boundary. Initially there is no wave in the region $x < 0$ because the initial data are supported in $(0, 1)$. Therefore, only waves leaving the computational domain toward the left are allowed in the exterior region.

Such an outgoing wave has the form

$$u(t, x) = \psi(x + t).$$

Differentiating,

$$u_t = \psi'(x + t), \quad u_x = \psi'(x + t),$$

which gives

$$u_t - u_x = 0.$$

Hence, the exact non-reflecting boundary condition at the left boundary is

$$\boxed{(\partial_t - \partial_x) u(t, 0) = 0.}$$

This condition eliminates any wave entering the computational domain from the left.

Transparent boundary condition at right boundary. Similarly, in the exterior region $x > 1$, only waves travelling to the right are allowed. Such waves have the form

$$u(t, x) = \phi(x - t).$$

Differentiating,

$$u_t = -\phi'(x - t), \quad u_x = \phi'(x - t),$$

so that

$$u_t + u_x = 0.$$

Therefore, the transparent boundary condition at the right boundary is

$$\boxed{(\partial_t + \partial_x) u(t, 1) = 0.}$$

This condition prevents any wave from entering the computational domain from the right.

Equivalent initial-boundary value problem. The original Cauchy problem on $\mathbb{R}$ is therefore equivalent to the following problem posed on the bounded interval $(0, 1)$:

$$\begin{cases} \partial_{tt}u - \partial_{xx}u = F(t, x), & 0 < x < 1, t > 0, \\ u(0, x) = f(x), \\ \partial_t u(0, x) = g(x), \\ (\partial_t - \partial_x)u(t, 0) = 0, \\ (\partial_t + \partial_x)u(t, 1) = 0. \end{cases}$$

These boundary conditions are called the *transparent*, *absorbing*, or *non-reflecting* boundary conditions, since they allow waves generated inside the interval $(0, 1)$ to leave the computational domain without producing artificial reflections.

REFERENCES

- [1] Dietrich Braess. *Finite Elemente - Theorie, schnelle Löser und Anwendungen in der Elastizitätstheorie*. Berlin [u.a.]: Springer, 1992. ISBN: 3-540-54794-0.
- [2] Susanne C. Brenner and Larkin R. Scott. *The Mathematical Theory of Finite Element Methods. Vol. 15. Texts in Applied Mathematics*. Springer, 2008. isbn: 978-0-387-75933-3. doi: 10.1007/978-0-387-75934-0. URL: <http://dx.doi.org/10.1007/978-0-387-75934-0>.
- [3] K. Eriksson. *Computational Differential Equations*. Cambridge University Press, 1996. ISBN:978-0-521-56738-1. URL: <http://www.worldcat.org/isbn/9780521567381>.
- [4] B. Gustafsson, H.O. Kreiss, and J. Oliger. *Time Dependent Problems and Difference Methods*. A Wiley-Interscience Publication. Wiley, 1995. ISBN: 978-0-471-50734-5. URL: <https://books.google.ch/books?id=1JZ2mSQ-06MC>.
- [5] F. Ihlenburg. *Finite Element Analysis of Acoustic Scattering. Applied Mathematical Sciences*. Springer, 1998. ISBN: 978-0-387-98319-6. URL: <https://books.google.ch/books?id=HcdwQ7heAG8C>.
- [6] G. Strang and G. Fix. *An Analysis of the Finite Element Method*. Wellesley-Cambridge Press, 2008. ISBN: 978-0-9802327-0-7. URL: <https://books.google.ch/books?id=K5MA0wAACAAJ>.